From Sensitization to Molecular Mechanism: Why AI Must Complement Experimental IgE Epitope Mapping

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Abstract

Allergen-specific immunoglobulin E (IgE) testing identifies sensitization but does not, by itself, establish clinical reactivity or explain the molecular basis of allergen recognition. This literature review examines how successive experimental and computational technologies narrow that inferential gap, from component-resolved diagnostics and basophil activation testing to single-cell antibody recovery, human monoclonal IgE, structural epitope mapping, and artificial intelligence (AI). Nineteen peer-reviewed sources were synthesized according to the type of claim each method can support, the evidence used for validation, and the uncertainty that remains. The reviewed evidence shows that each technology answers a distinct question: molecular assays identify recognized allergen components; functional assays test effector-cell activation; single-cell methods identify antibody clones; structural approaches localize antibody-allergen interfaces; and clinical studies establish patient-level relevance. AI and protein-structure prediction offer substantial value by prioritizing candidate allergens, structures, and epitope regions at scales that are impractical for exhaustive experimental testing. However, strong performance in whole-protein allergen classification does not establish accurate residue-level IgE epitope prediction, and computationally plausible surface regions cannot be considered validated epitopes without experimental confirmation. The most defensible approach is therefore complementary rather than substitutive: AI should reduce the experimental search space, while binding assays, mutagenesis, cellular activation studies, structural methods, and clinical evidence test whether predicted interactions are biologically and clinically meaningful. Integrating these layers creates an evidence chain that moves from sensitization toward experimentally validated molecular mechanisms and, ultimately, clinically actionable

interpretation. This framework also clarifies a key limitation in current allergy research: model performance must be interpreted at the same biological level at which the model was trained and validated, rather than extrapolated across distinct diagnostic, molecular, or clinical tasks.

Keywords: immunoglobulin E; allergy diagnostics; component-resolved diagnostics; basophil activation test; single-cell sequencing; human monoclonal IgE; conformational epitopes; epitope mapping; artificial intelligence; allergen structure

Introduction

Historically, allergy testing centered on whether serum contained allergen-specific IgE. That question remains clinically useful, but it is narrower than the biology of disease: receptor-bound IgE must cross-link on mast cells or basophils to produce symptoms, and a positive serum result can reflect sensitization without symptomatic reactivity. The central problem is therefore inferential. Simply detecting IgE ignores whether cells will respond, which rare B-cell clones generated the antibodies, which molecular surface permits binding, or whether disrupting that mechanism will protect patients (Ansotegui et al., 2020).

Conceptual synthesis of IgE-mediated effector-cell activation and the distinct questions answered by serum and multiplex assays, BAT, single-cell methods, structural biology, AI, and clinical trials (Ansotegui et al., 2020; Doña et al., 2021; Pomés et al., 2024).

The first response to this limitation was greater molecular specificity. Extract-based tests

can obscure the responsible component or cross-reactive family, so component-resolved and multiplex assays shifted the field from whole-extract detection to component profiles. Yet a more detailed profile still records recognition, not cellular activity, and platform-specific thresholds can complicate interpretation. The unresolved question of whether detected IgE can activate effector cells motivates functional assays such as the basophil activation test (Ansotegui et al., 2020; Doña et al., 2021).

BAT narrows the gap between sensitization and reaction, but a cellular response still does not identify the antibody clones that produced it. Since IgE-producing B cells are rare and serum is polyclonal, single-cell sequencing is needed to recover paired antibody sequences. Patient-derived monoclonal IgE can then convert those clones into renewable reagents for functional testing and structural mapping. Structural methods localize antibody-allergen interfaces, but remain too intensive to apply across every possible clone-allergen pair. (Croote et al., 2018; Pomés et al., 2024).

The strongest rationale for computation is the issue of scaling. AI can search large protein, structure, and antibody spaces faster than experimental mapping, but speed does not erase biological distinctions. DeepAlgPro classifies whole proteins, whereas residue-level prediction asks which amino acids human IgE recognizes. Their unequal performance reveals the central methodological problem: computational output must be interpreted at the level it was trained and validated to predict (He et al., 2023; Kwon et al., 2024).

This literature review determines how increasingly high-resolution experimental and computational approaches change what researchers can infer about IgE epitopes, and where

AI-generated predictions must be experimentally validated. The hypothesis of this study is that each technology addresses distinct inferential limitations left by prior approaches, therefore no single method can support the full progression from sensitization to clinically actionable mechanisms. Rather than treating newer technologies as replacements, we evaluate the evidence chain required to move from detection to functional, molecular, computational, and clinical claims.

Methods

This focused narrative review included 19 core peer-reviewed sources selected for direct relevance to IgE detection, functional testing, single-cell and repertoire analysis, structural epitope mapping, AI-assisted prediction, or therapeutic targeting. Searches covered PubMed, PubMed Central, Google Scholar, ScienceDirect, and publisher databases. Search terms combined IgE with allergy diagnostics, component-resolved diagnostics, multiplex IgE, basophil activation test, single-cell RNA sequencing, V(D)J sequencing, IgE repertoire, human IgE monoclonal antibody, conformational epitope, allergen structure, AlphaFold, allergen prediction, machine learning, epitope prediction, omalizumab, and therapeutic monoclonal antibody. This paper considers only original research articles that quantitatively describe some aspect of protein-protein interaction performance, involve sample size determination, assess the structure of the molecule, or present clinical consequences. Rather than organizing evidence by the

chronological development of the technology, we structure our discussion around the inferential logic behind the use of PPIs: detectability, functional role, clone identification, interface identification, prioritizing computationally, validating experiments, and clinical significance. Our comparisons consider the unit of analysis, goal of validation, and remaining uncertainties. Position papers and critical reviews provide contextual information from disciplines and clinical perspectives; primary research provides quantitative and methodological information.

Results

Detection Identifies Sensitization, Not Clinical Reactivity

Measurement of serum allergen-specific IgE provides an affirmative response to the first question in this chain: Is there any evidence that the individual has generated an antibody response against a particular allergen? As stated by Ansotegui et al. (2020), in its position paper on this topic, positive results demonstrate sensitization but do not demonstrate allergy. Positive results cannot provide any indication that the individual would suffer from symptoms caused by activation of the effector response through crosslinking of enough IgE.

Component-resolved diagnostics replace crude extracts with purified molecules, improving

molecular specificity. In a comparison of ALEX2 and ISAC112, Scala et al. (2021) report a 94.3% agreement across shared molecules but find only 71% after double-negative results were removed and quantitative values transferred poorly. Thus, component profiling identifies potentially relevant allergens, but platform agreement does not establish biological activity; that requires a functional test.

Functional Assays Test Activity, Not Clonal Identity

BAT supplies that next layer by exposing a patient's basophils to a suspected allergen and measuring activation markers such as CD63 or CD203c. Doña et al. (2021) position BAT as a complementary test when history, skin testing, and serum IgE disagree or when oral challenge is risky. Its contribution is conceptual as well as practical: it asks whether the patient's cell-bound IgE can produce a cellular response.

In peanut-sensitized children, Santos et al. (2014) reported 97% diagnostic accuracy and showed that a BAT-based strategy could reduce oral food challenges by two-thirds. Those results support functional discrimination in that cohort; they do not establish a universal cutoff. BAT still depends on rapid sample processing, laboratory protocol, allergen, and population, so local validation remains necessary.

The assay demonstrates any change in signaling due to an intervention under consideration. For example, El Ansari et al. (2022) discovered that allergen-specific IgA inhibited IgE-mediated degranulation of mast cells, syk-signaling pathways, and basophil

activation by peanuts. The assay goes one step further in demonstrating efficacy after binding events. Still, it does not solve the issue at the population level of identifying the individual IgE clones producing the signal, nor can it produce them for further examination.

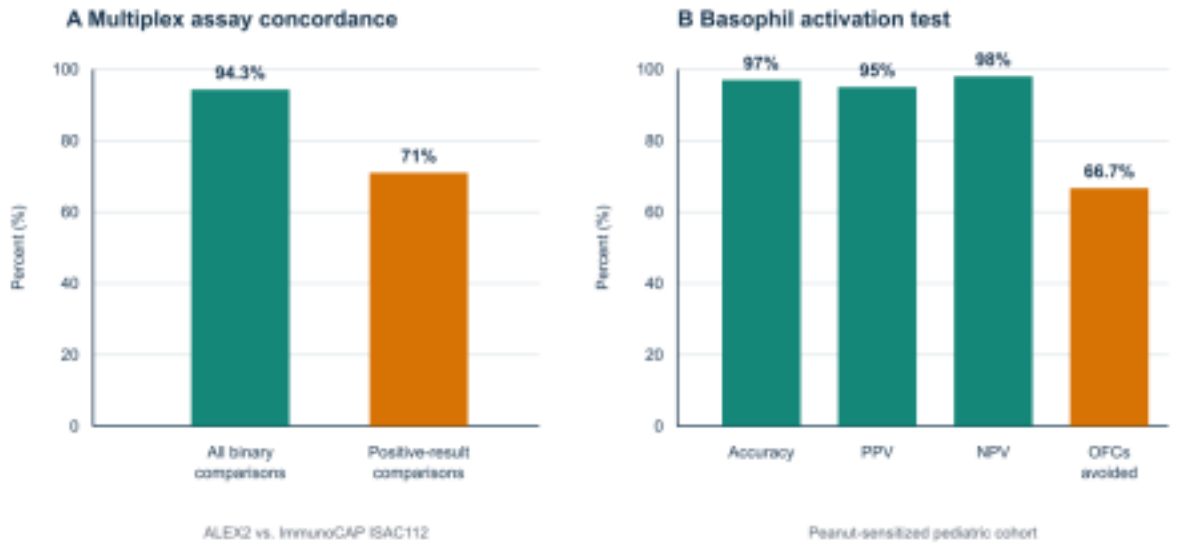


Figure 1

Selected diagnostic performance of multiplex and basophil activation testing Figure 2. Panel A compares binary concordance across all allergen results with concordance after excluding double-negative results (Scala et al., 2021). Panel B reports BAT accuracy, positive predictive value (PPV), negative predictive value (NPV), and the proportion of oral food challenges (OFCs) avoided in a peanut-sensitized pediatric cohort (Santos et al., 2014).

Single-Cell Methods Identify Clones, Not Molecular Interfaces

Single-cell methods reveal clonal identity hidden within polyclonal serum. Croote et al. (2018) recovered paired antibody sequences from single IgE B cells in food-allergic subjects and

linked somatic hypermutation to high-affinity, cross-reactive antibodies against Ara h 2 and Ara h 3. These methods connect a biological response to an antibody lineage, but do not by themselves provide a renewable reagent.

Human monoclonal IgE makes identified antibody lineages experimentally reusable. Patient-derived antibodies enabled atomic mapping of2F10 IgE epitope on Der p 2, while Smith

et al. (2023) developed 33 monoclonal antibodies from 17 patients. Renewable clones can be tested repeatedly across binding, signaling, and structural assays directly linking clonality to function.

(Smith et al., 2023).

Cell sorting can bias which clones are recovered. Thörnqvist et al. (2026) addressed this limitation with an allergen-agnostic pipeline combining single-cell V(D)J sequencing, repertoire matching, recombinant expression, and mass spectrometry, recovering high-affinity IgE to four grass-pollen allergens. The approach broadens clone recovery but remains limited by small cohorts and complex downstream processing.

Structural Mapping Resolves Interfaces but Does Not Scale

Epitope mapping addresses where antibody-allergen binding occurs. Because many IgE

epitopes depend on three-dimensional structure Pomés et al. (2024) emphasize a validation sequence; localize a candidate interface, mutate key residues, and test whether binding or biological activity changes.

Structural resolution matters because an interface can explain intervention. Pennington et al. (2016) showed how the omalizumab–IgE-Fc structure accounts for blockade of FcεRI and CD23 interactions and guided an engineered IgE-Fc variant that enhanced inhibition in a basophil system when combined with omalizumab. This was mechanistic proof of concept, not a clinical efficacy result. It demonstrates the value of structure while preserving the distinction between molecular design, cellular validation, and patient benefit.

Experimental structural data remain costly and unavailable for most IgE-allergen interactions (Pomés et al., 2024). AlphaFold showed that protein-fold prediction can be highly accurate at scale (Jumper et al., 2021), allowing predicted allergen structures to prioritize exposed regions for testing. However, a correct allergen fold does not prove an IgE-binding interface because flexibility, glycosylation, oligomerization, and antibody-specific geometry can alter binding.

AI Prioritizes Candidates but Does Not Validate Epitopes

Comprehensive experimental structural characterization is impractical across the vast space of proteins, epitopes, antibodies and patients (Pomés et al., 2024). Computational methods address this scale, but their outputs must be interpreted only at the task on which each model was trained and validated, such as structure prediction, allergen classification, epitope prediction, or

clinical outcome prediction. Treating success on one task as evidence for another creates biological claims that were never experimentally tested. (Jumper et al., 2021; He et al., 2023; Kwon et al., 2024).

AI allergen prediction has progressed from sequence-similarity methods to deep-learning classifiers. ALLERDET combines sequence alignment with machine-learning models, whereas DeepAlgPro uses convolution and attention across full protein sequences. Both answer a protein-level question: whether a protein is likely to be allergenic. DeepAlgPro shows promise due to its capacity for efficient allergen triaging and identification of new allergens, which makes the technique suitable for applications requiring rapid screening of large protein datasets and prioritization of candidates for downstream experimental investigation. However, assigning an allergenicity label to an entire protein does not identify the specific residues or three-dimensional surface recognized by human IgE. Strong performance in protein-level classification therefore cannot be interpreted as evidence of accurate IgE epitope localization, which requires separate residue-level models and experimental validation.

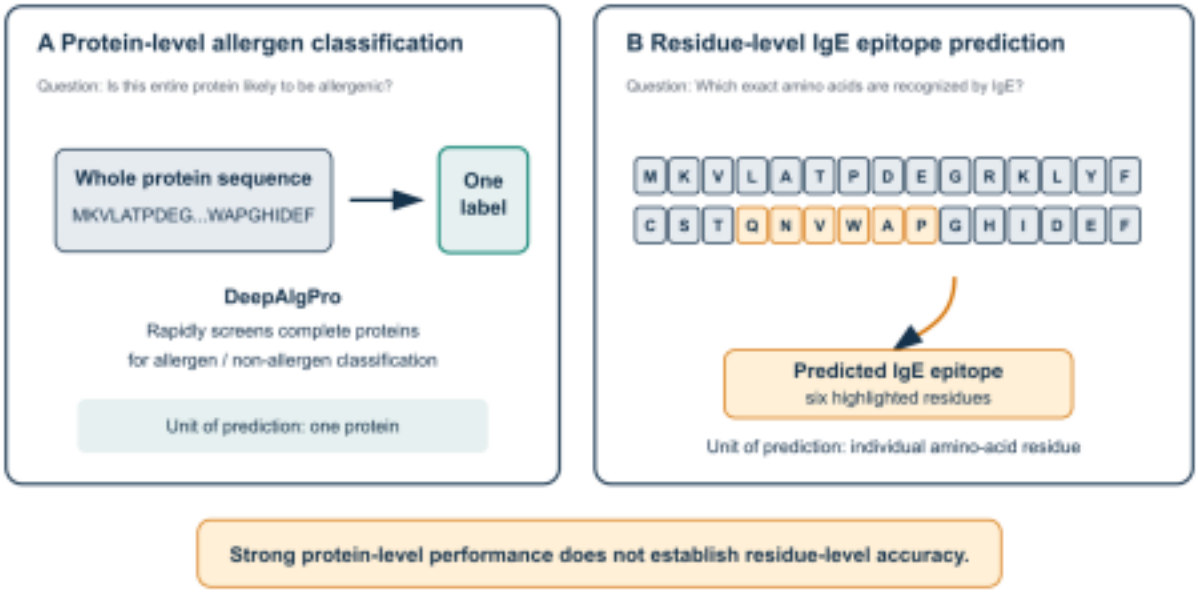


Figure 2

AI Image of Difference between protein-level allergen classification and residue-level IgE epitope prediction

Protein-level systems such as DeepAlgPro assign one label to an entire protein, whereas residue-level systems attempt to identify the specific amino acids forming an IgE-binding region. These tasks use different labels and validation standards; strong protein-level performance does not establish residue-level accuracy (He et al., 2023; Kwon et al., 2024).

Residue-level IgE prediction asks which amino acids participate in recognition. Kwon et al. (2024) evaluated nine public tools on Fel d 1 and across 59 allergens; most Matthews correlation coefficients were near zero, with non-epitopes predicted more reliably than epitopes. This does not contradict strong protein classifiers; it highlights the gap between classifying an allergen and identifying the residues bound by human IgE.

Experimental follow-up is therefore part of the AI claim. Yu et al. (2023) used machine learning to prioritize beta-lactoglobulin peptide regions and ELISA to support residues 116-130 as an IgE-binding epitope. The study connects prediction to measurement, while also showing that peptide-level validation does not establish a conformational epitope or clinical reaction. Related AI work in vaccine development supports the scalability of candidate prioritization but is not IgE-specific validation (Olawade et al., 2024). Allergy models ultimately require datasets linking sequence and structure to paired human antibodies, binding, effector-cell activation, and patient phenotype.

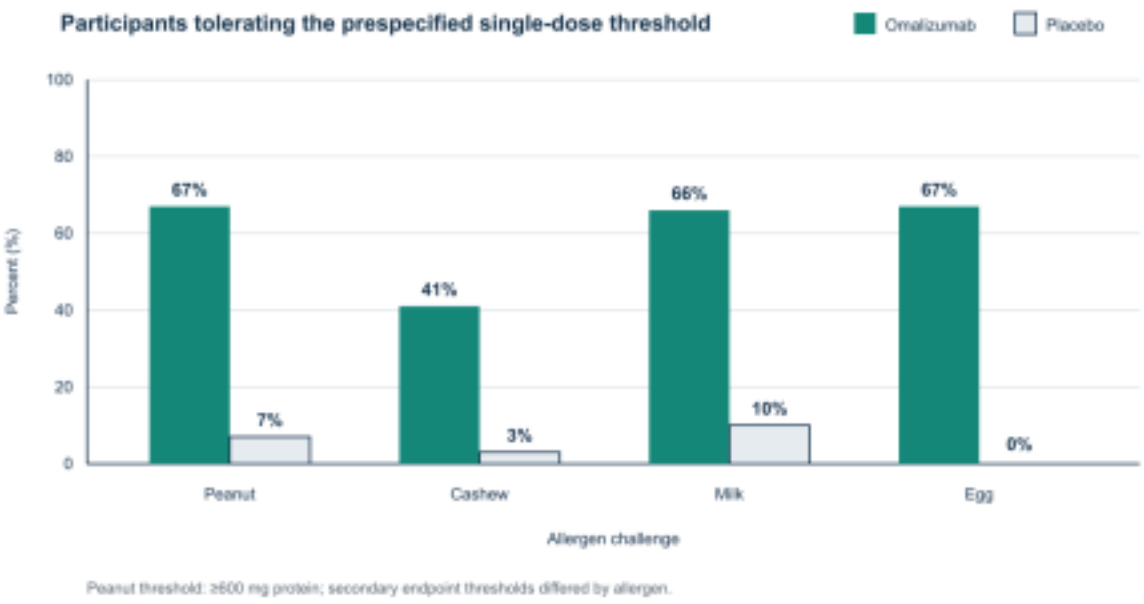
Clinical Trials Establish Patient Relevance

Even a validated molecular mechanism does not establish patient benefit. Eggel et al. (2024) review therapies directed at IgE, downstream cytokines, and epithelial alarmins, showing that mechanistic knowledge can identify multiple intervention points. The remaining question is clinical: does altering the mechanism raise reaction thresholds, reduce risk, or improve outcomes in patients?

OUtMATCH provides that patient-level test for anti-IgE therapy. After 16–20 weeks, 67% of participants receiving omalizumab tolerated at least 600 mg of peanut protein, compared with 7% receiving placebo; benefits were also observed for other prespecified allergens (Wood et al., 2024). The primary comparison demonstrates increased protection against accidental exposure, not cure. Continued avoidance and ongoing dosing remained necessary, so the trial completes the evidentiary chain while defining the limit of the intervention.

Figure 3

OUtMATCH response rates by allergen and treatment group



Note. Bars show the percentage of participants who tolerated the prespecified single-dose threshold after 16–20 weeks. Peanut threshold: at least 600 mg protein; secondary endpoint thresholds differed by allergen. Source: Wood et al. (2024).

Taken together, the technologies form an inference ladder rather than a sequence of replacements. Detection establishes sensitization; functional assays establish cellular activity; single-cell methods identify responsible clones; structural mapping localizes interfaces; AI prioritizes candidates; experiments validate binding or function; and clinical research establishes patient relevance. Each layer is necessary because the preceding one leaves a different claim unsupported.

Evidence by Layer: What is Proven vs. Unresolved

Technology	Representative quantitative evidence	Interpretation and limitation
Multiplex CRD	ALEX2 vs. ISAC112: 140 patients; 94.3% binary concordance, falling to 71% without double negatives (Scala et al., 2021).	Broadens molecular profiling. Positive and quantitative results remain platform dependent.
Basophil activation test	Peanut BAT: 97% accuracy, 95% PPV, 98% NPV; two-thirds fewer OFCs (Santos et al., 2014).	Tests effector-cell function. Requires rapid processing and allergen-specific validation.
Single-cell and repertoire	Thirty-three renewable human IgE monoclonals from 17 patients; allergen-agnostic pipeline recovered IgE to four grass allergens (B. R. E. Smith et al., 2023; Thörnqvist et al., 2026).	Links clones to specificity. Cohorts and recovered repertoires remain small.
Human IgE structural mapping	The 2F10 human IgE epitope covered approximately 750 Å² on Der p 2 (S. A. Smith et al., 2023).	Provides atomic mechanisms, but few patient-derived IgE complexes are available.

Technology	Representative quantitative evidence	Interpretation and limitation
Structure prediction	AlphaFold CASP14 median RMSD95: 0.96 Å vs. 2.8 Å for the next-best method (Jumper et al., 2021).	Supplies structural hypotheses, not proof of IgE binding or clinical reactivity.
Protein allergen classification	ALLERDET used sequence alignment with an RBM/decision tree; DeepAlgPro recovered 21–22 of 24 reported novel allergens and screened 1,000 proteins in 0.287 s (Garcia-Moreno & Gutiérrez-Naranjo, 2022; He et al., 2023).	Rapid protein-level triage. Does not localize conformational IgE epitopes.
AI epitope prediction	Nine public tools tested across 59 allergens; most MCC values were near zero (Kwon et al., 2024).	Shows residue-level generalization, but weak and task-specific.
Anti-IgE therapy	OUtMATCH: 67% vs. 7% tolerated ≥600 mg peanut protein; cashew 41% vs. 3%, milk 66% vs. 10%, egg 67% vs. 0% (Wood et al., 2024).	Demonstrates clinical protection, but not a cure; avoidance and ongoing dosing remain necessary.

Table 1: *Quantitative evidence, contribution, and limits across IgE technologies*

Note: BAT = basophil activation test; CRD = component-resolved diagnostics; MCC = Matthews correlation coefficient; OFC = oral food challenge; RMSD95 = root-mean-square deviation calculated over 95% of residues.

Discussion

The central finding is that technical resolution and inferential strength are not the same. Serum, functional, molecular, computational, and clinical methods validate different objects under different conditions. Their evidence becomes cumulative only when the output of one level is explicitly tested at the next.

Quantitative results must therefore be interpreted at their own level. Multiplex concordance measures agreement between platforms; BAT accuracy measures diagnostic discrimination; DeepAlgPro evaluates whole-protein classification; Kwon et al. test residue localization; and OUtMATCH measures patient tolerance after treatment. These outcomes are not directly rankable; together, they show which inference is supported and what still requires another method.

Clinical usefulness is the strictest test. Future datasets should distinguish allergen classification, antibody binding, effector-cell activation, and clinical reactivity while linking them across the same patients and antibody clones. External cohorts, prospective validation, calibrated uncertainty, and diverse populations are needed to prevent task-specific

performance from being mistaken for generalization. The goal is reproducible workflow in which computation

reduces experimental burden while experimental and clinical evidence constrain. Because this was a focused narrative review rather than a systematic review, its conclusions should be interpreted as a structured synthesis rather than an exhaustive or directly quantitative comparison.

Conclusion

IgE research has progressed from measuring sensitization to testing cellular activity, identifying responsible clones, localizing binding interfaces, and determining clinical relevance. AI is most useful when molecular search spaces become too large for exhaustive experimentation, but its predictions remain hypotheses rather than validated epitopes or clinical conclusions. The strongest workflow is iterative: computation narrows the search space, human IgE reagents and functional assays test the prediction, structural methods explain the interaction, and clinical studies establish patient relevance. . The key advance is therefore an integrated evidence chain that moves from sensitization forward toward clinically actionable mechanisms.

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